UNIT
1

UNDERSTANDING GENERATIONS OF COMPUTER

Learning Objectives

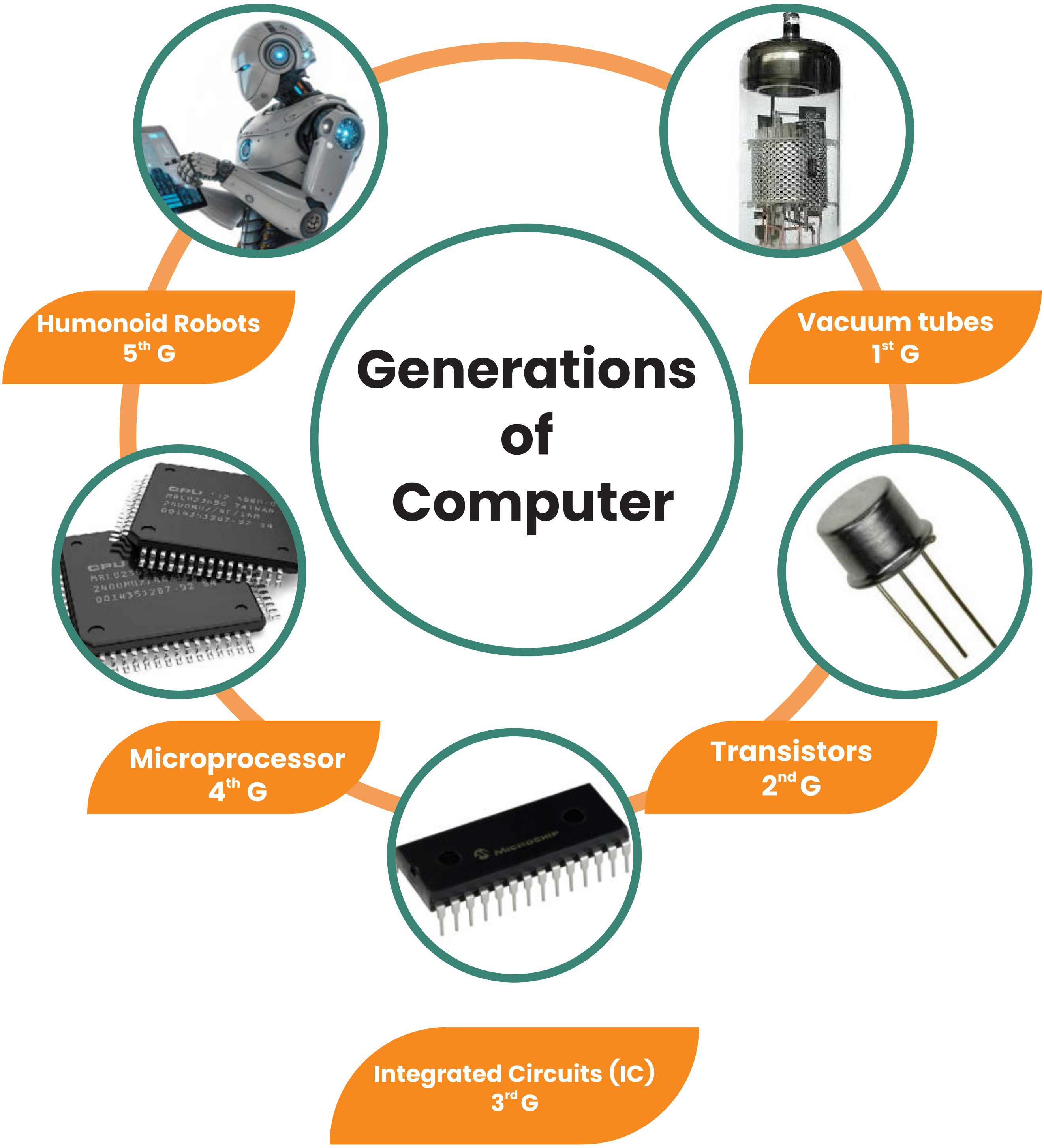
- Define computer generations and their significance.
- Identify the five generations of computers and their key technologies.
- Describe the major advancements in hardware and software across generations.
- Compare the features and performance of different generations.
- Recognize examples of computers from each generation.
- Explain the impact of technological evolution on modern computing.
- Discuss future trends based on past developments.

Generations of Computers

Generation in computer terminology refers to the evolution of computer technology over time. Initially, the term was used to distinguish between varying hardware technologies. Today, it encompasses both hardware and software advancements that define entire computer systems.

There are five recognized generations of computers, each marked by significant technological developments. The following table outlines these generations along with their approximate time periods and key characteristics:

Generation	Time Period	Key Technology
First Generation	1946-1959	Vacuum tube-based
Second Generation	1959-1965	Transistor-based
Third Generation	1965-1971	Integrated Circuit-based
Fourth Generation	1971-1980	VLSI microprocessor-based
Fifth Generation	1980-present	ULSI microprocessor-based


Generations of Computer
Vacuum tubes
1st G
Transistors
2nd G
Integrated Circuits (IC)
3rd G
Microprocessor
4th G
Humonoid Robots
5th G

Key Electronic Components and Their Roles

Vacuum Tubes

An electronic device that controls the flow of electrons in a vacuum. It was used as a switch, amplifier, or display screen in many older model radios, televisions, computers, etc

Artificial Intelligence (AI)

An area of computer science that deals with the simulation and creation of intelligent machines (they think, learn, work, and react like humans).

Transistor

An electronic component that can be used as an amplifier or as a switch. It is used to control the flow of electricity in radios, televisions, computers, etc.

Microprocessor

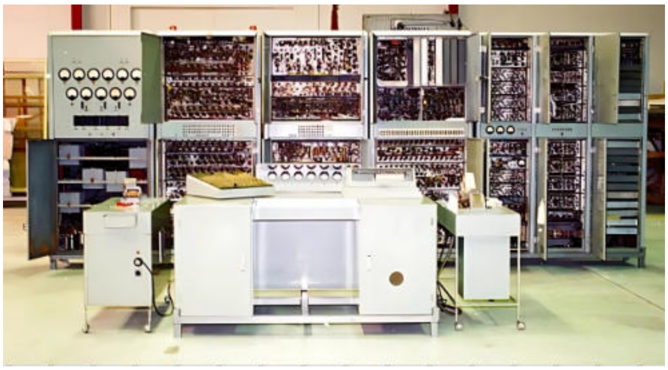
An electronic component held on an integrated circuit that contains a computer's central processing unit (CPU) and other associated circuits.

Integrated circuits (IC)

A small electronic circuit printed on a chip (usually made of silicon) that contains its own circuit elements (e.g. transistors, diodes, resistors, etc.).

1. First Generation Computers (1946-1959)

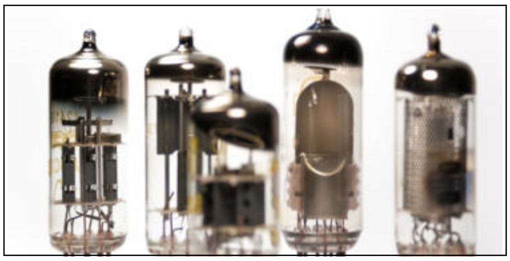
Computers of the first generation used vacuum tubes for memory and CPU circuitry. These machines generated excessive heat and required frequent maintenance. Due to their high cost, only large organizations could afford them. They relied on batch processing operating systems, with punch cards and magnetic tapes as input and output devices. Machine language was the only programming language used.



1ST First Generation computer

Features:

- Vacuum tube technology
- Unreliable and expensive
- Machine language support only
- Large size and high electricity consumption
- Required air conditioning



Vaccum tubes

Examples:

- ENIAC
- EDVAC
- UNIVAC
- IBM-701
- IBM-750

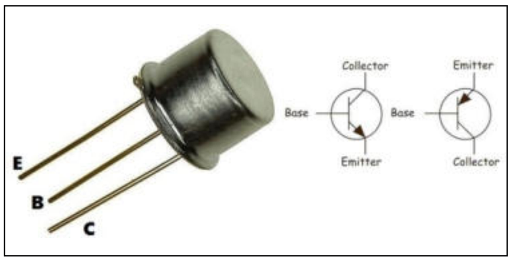
2. Second Generation Computers (1959-1965)

This generation introduced transistors, which made computers smaller, faster, and more reliable. Magnetic cores replaced vacuum tubes as primary memory, while secondary storage used magnetic tapes and disks. Assembly language and high-level programming languages like FORTRAN and COBOL were introduced.

2nd **Generation** computer

Features:

- Transistor technology
- More reliable and compact than first-generation computers
- Lower heat production and power consumption
- Faster processing speed
- Still expensive but improved affordability



Transistors

Examples:

- IBM 1620
- IBM 7094
- CDC 1604
- UNIVAC 1108

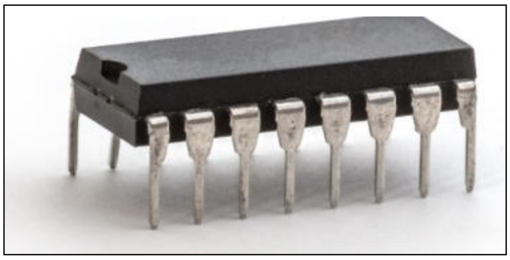
3.Third Generation Computers (1965-1971)

This generation saw the development of Integrated Circuits (ICs), invented by Jack Kilby. Computers became smaller, more efficient, and reliable. Time-sharing and multiprogramming operating systems emerged, and high-level programming languages like BASIC, PASCAL, and ALGOL were widely used.

3rd Generation computer

Features:

- IC technology
- Smaller and more efficient
- High-level language support
- Reduced heat generation
- Faster processing



Integrated circuit (IC)

Examples:

- IBM-360 series
- Honeywell-6000 series
- PDP
- IBM-370/168

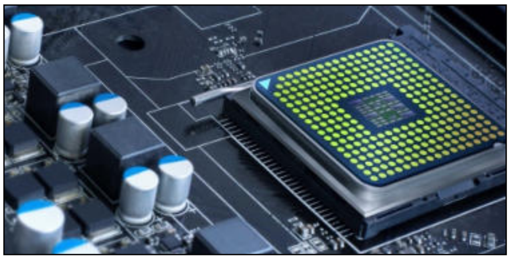
4. Fourth Generation Computers (1971-1980)

With the advent of Very Large Scale Integration (VLSI), thousands of transistors could be integrated into a single chip. This led to the rise of microcomputers and personal computers. The internet concept was introduced, and operating systems became more advanced with real-time processing and distributed computing.

4^{th} **Generation** computer

Features:

- VLSI technology
- Personal computers became widespread
- Highly portable and affordable
- Internet and networking concepts introduced



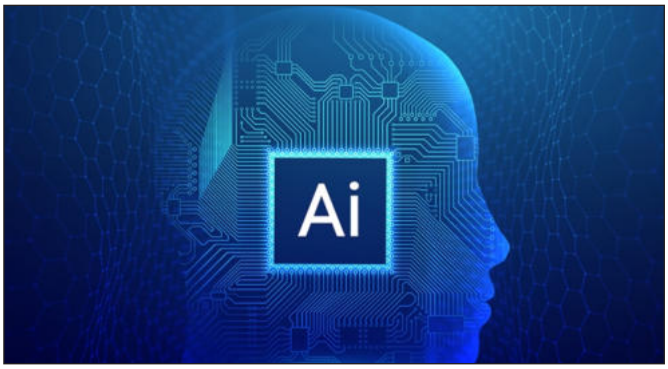
Microprocessor

Examples:

- DEC 10
- STAR 1000
- PDP 11
- CRAY-1 (Super Computer)

5. Fifth Generation Computers (1980-Present)

Fifth-generation computers use Ultra Large Scale Integration (ULSI) technology. They are characterized by artificial intelligence (AI), natural language processing, parallel processing, and advanced multimedia capabilities. This generation includes desktops, laptops, notebooks, ultrabooks, and cloud computing systems.



5TH Generation computer

Features:

- ULSI technology
- Artificial intelligence and machine learning advancements
- Parallel processing capabilities
- User-friendly interfaces
- More powerful yet compact computers at affordable prices

Examples:

- Desktop computers
- Laptops
- Notebooks
- Ultrabooks
- Chromebooks

Group Discussion:

✓Debate the impact of AI in fifth-generation computers.

EXERCISES

01: Choose the correct answer.

1. Which technology was used in first-generation computers?

a) Transistors

b) Vacuum tubes

c) Integrated circuits

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2. What was the major improvement in second-generation computers?

a) Use of microprocessors

b) Introduction of transistors

c) Artificial intelligence

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3. Which key technology was introduced in the third-generation computers?

a) Vacuum tubes

b) Integrated Circuits (ICs)

c) Cloud computing

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4. The fourth generation of computers saw the rise of:

a) Personal computers

b) Punch cards

c) Magnetic tapes

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5. Fifth-generation computers are mainly characterized by:

a) Artificial intelligence and ULSI technology

b) Vacuum tubes and transistors

c) Batch processing

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6. The first generation of computers was developed in which time period?

a) 1959-1965

b) 1971-1980

c) 1946-1959

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7. Which programming languages were introduced in the second generation?

a) FORTRAN and COBOL

b) C++ and Java

c) Python and Swift

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8. What made fourth-generation computers more efficient?

a) Punch cards

b) VLSI technology

c) Magnetic drums

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9. The major difference between third and fourth-generation computers is:

a) Use of transistors instead of vacuum tubes

b) Introduction of microprocessors

c) Development of AI

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10. Which of the following is an example of a fifth-generation computer?

a) IBM-701

b) ENIAC

c) Laptop

02: Fill in the blanks.

1. The first generation of computers used ___________ technology.

2. Transistors replaced vacuum tubes in the ___________ generation.

3. The third generation of computers introduced ___________ circuits.

4. The development of microprocessors began in the _________ generation.

5. ___________ is a characteristic of fifth-generation computers.

6. The language used in first-generation computers was ___________.

7. The fourth-generation computers led to the development of ___________.

8. ___________ and ___________ are examples of third-generation computers.

03: Select 'True' or 'False'.

	TRUE	FALSE
1. The first-generation computers used transistors.		
2. COBOL and FORTRAN were introduced in the second generation.		
3. The fourth generation of computers introduced vacuum tubes.		
4. Supercomputers are best suited for simple arithmetic operations.		
5. Third-generation computers used Integrated Circuits (ICs).		
6.Second-generation computers were faster and more reliable than first-generation computers.		
7. The main storage device in first-generation computers was a solid-state drive.		
8. ULSI technology is used in the fourth-generation computers.		

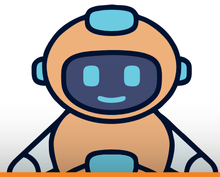
04: Answer the following questions briefly:

1. What are computer generations?
2. How did transistors improve computers in the second generation?
3. What is the main advantage of Integrated Circuits (ICs)?
4. How did the introduction of microprocessors change computing?
5. Mention two key features of fourth-generation computers.
6. What are some modern applications of fifth-generation computers?

05: Provide detailed answers of the following questions:

1. What is computer generation? Discuss different generations of computers with the technologies used in each generation.
2. “The actual use and implementation of computers started after the third generation.” Justify this statement in your own words.
3. How has artificial intelligence influenced modern computers?
4. Discuss the role of microprocessors in computer advancement.

Lab Activities:



1. Comparing Technologies

–Create a presentation comparing different computer generations.

2. Hands-on with Old and New

– Find images of computers from different generations and categorize them accordingly.

3. Building a Timeline

– Design a timeline showcasing the key features and advancements of each generation.

4. Prepare a Short Report

Research and prepare a short report on how modern computers have evolved from the past.